Procurement Intelligence

Evidence-backed planning, tender design and monitoring for World Bank projects, built and measured on Databricks

Plan Tender design Monitor Learn

Technical assessment · Dhruv Agrawal · October 2026

Knowledge sits in three silos

STRUCTURED

What happened

Contract awards, projects, disbursements, notices

SEMI-STRUCTURED

How it was executed

ISRs, procurement plans, REOI / IFB notices

UNSTRUCTURED

Why it succeeded or failed

PADs, ICRs, IEG reviews, CPFs

The officer's journey: each step needs all three

1 Plan

How long, which method, what risk?

2 Tender design

Scope, criteria, lots: what went wrong before?

3 Monitor

Which tenders and projects need a look?

4 Learn

New reports update models and the index

One Databricks workspace, four modules

Open data

Contract awards (3 datasets)

Projects · notices

Disbursements

CPIA · WGI

PAD · ISR · ICR · IEG

M1 Ingest + knowledge

bronze → silver → gold

Quality gate: 14 / 14

Graph: nodes + edges

Parse once → chunks

Vector Search hybrid

M2 Models + RAG

Duration P20 / P50 / P80

Re-bid or cancellation

Ratings · lessons · events

Agentic RAG

Early warnings

M3 Guardrails

Citation · number · ID

Claim verifier

Injection quarantine

Conflicts: IEG > ICR > ISR

Users + M4 loop

App: Plan · Monitor · Ask

AI/BI KPI dashboard

Weekly job

Feedback → eval set

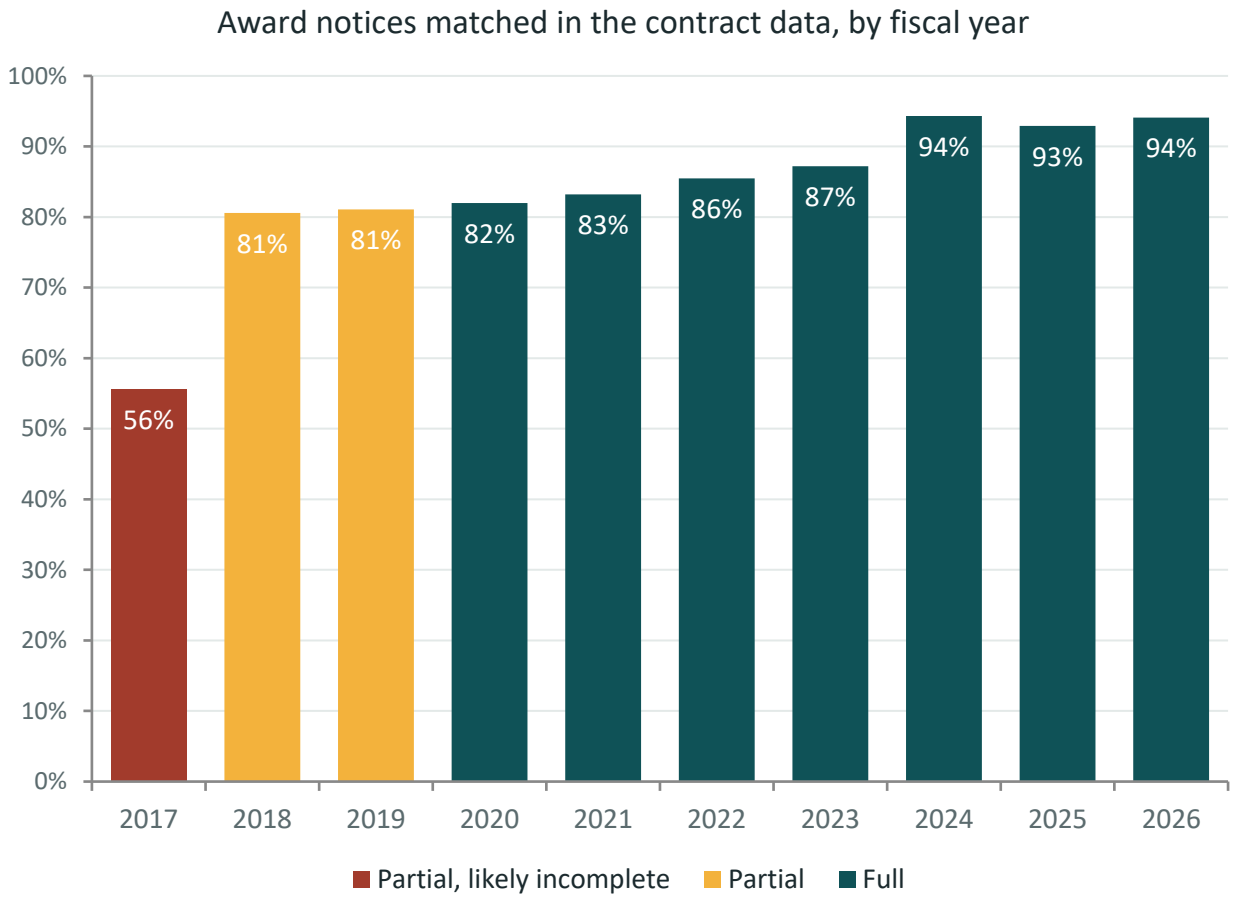
491,020
contracts

22,461
projects

9,564
indexed passages

16
demo projects, 5 countries

Data checked before any model was built



A third contract dataset

DS04691 covers FY2017–19: the assumed gap was partial coverage, not missing data

Plan dates rejected

Procurement-plan dates agreed with signing dates only 20% of the time

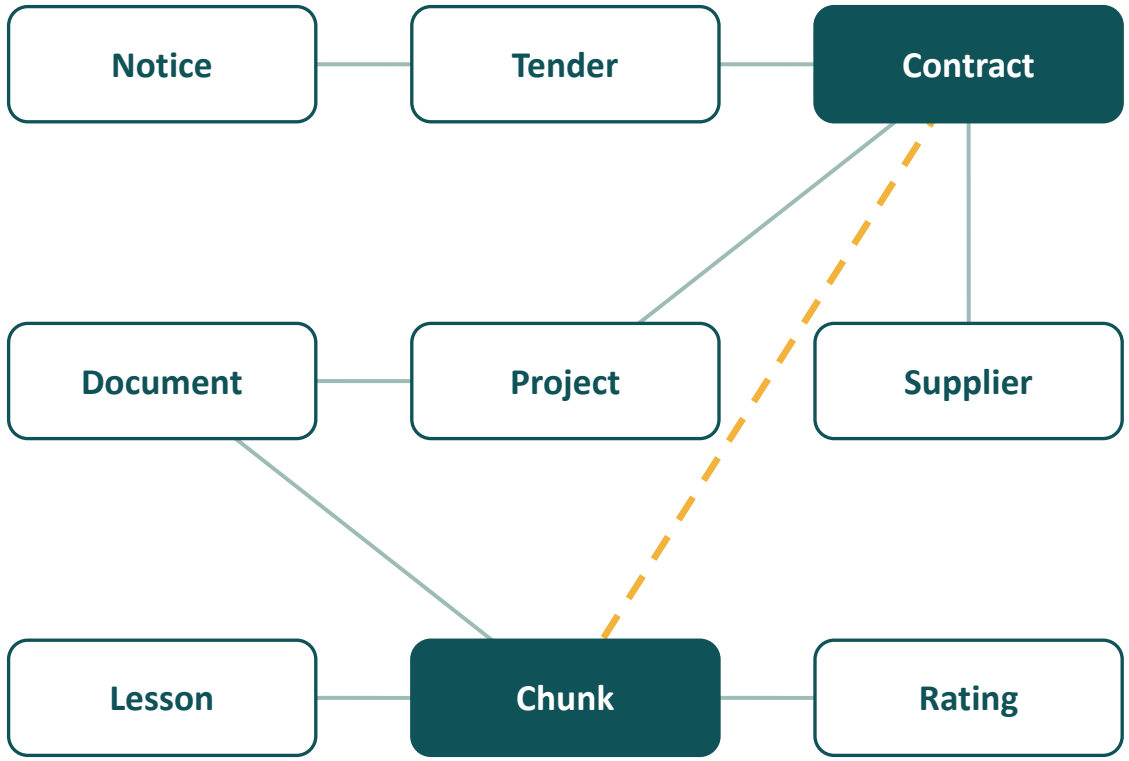
Target tested first

Post-review durations ≈13% negative, so the main model is prior-review

Automatic quality gate

14 checks, all pass; models do not run on a failed refresh

Every link carries its evidence and its date



DESCRIBES: a passage names the contract

1.3%

of demo contracts are named in any report passage (5 strong, 27 weak links)

So Vector Search is the primary path; links add “this passage names your contract” where they exist.

Traced on real data

Contract GYTJ201B → supplier → project → ISR passage reporting that this contract was re-bid

11 node types · 11 edge types · 0 orphan edges

Plan: how long will this tender take?



0.81

P80 coverage: the range holds for 81% of tenders (target 0.7–0.9)

≈ baseline

P50 error 100.4 vs 100.8 days: no sharper point estimate yet

94.7

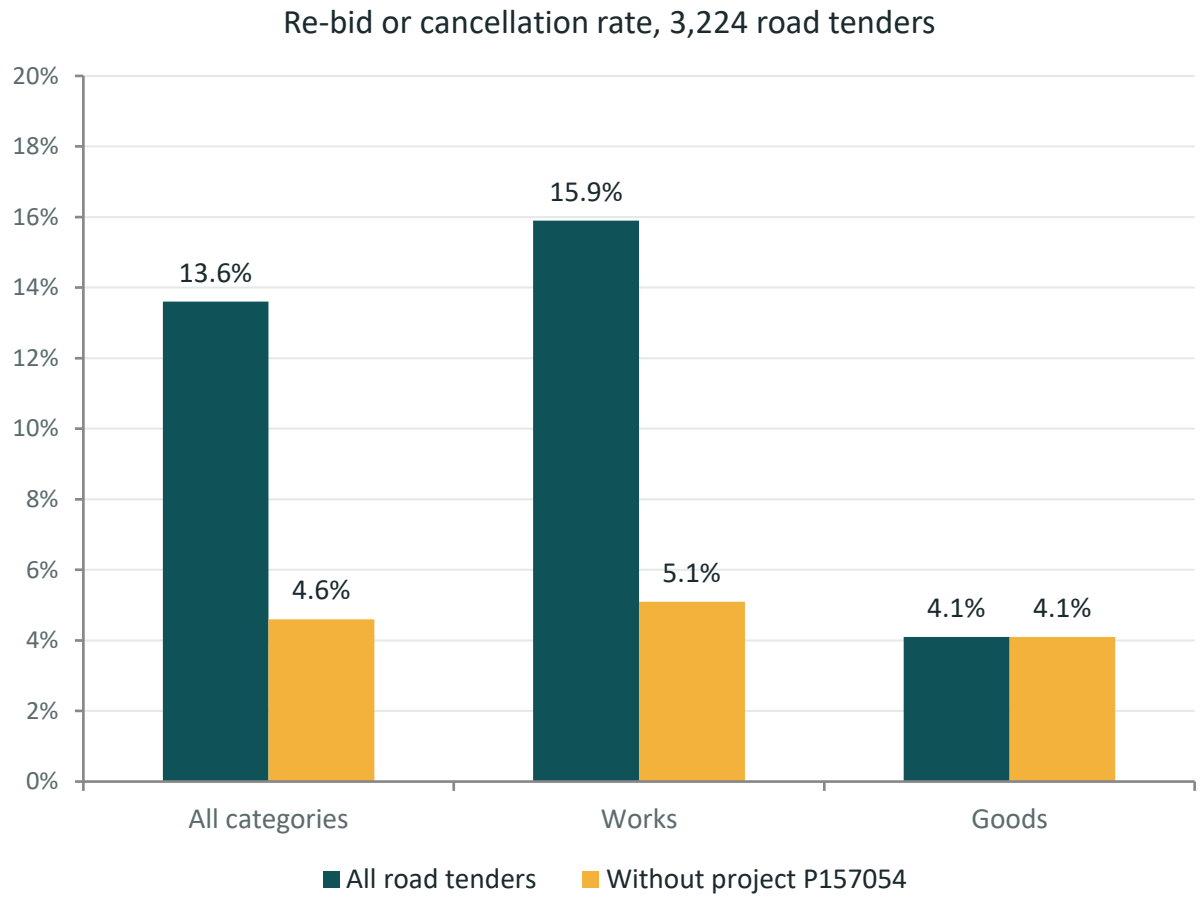
days MAE without governance features: WGI / CPIA did not help on 500 rows

Test set (time split)	n	P50 MAE (days)	Median error	P80 coverage
Prior review: median baseline	125	100.8	70.5	0.800
Prior review: quantile GBT (champion)	125	100.4	80.5	0.808
Post review: median baseline	519	70.3	57.5	0.827
Post review: quantile GBT	519	69.5	52.3	0.825

Features known at the start notice only · governance from the latest PUBLISHED year · promoted to champion only if no worse on the latest time slice

Plan: will it need re-bidding?

1 Plan — 2 Tender design — 3 Monitor — 4 Learn



97%

precision of the re-advertisement rule (29 of 30 sampled; 44 / 44 outside P157054)

0.032 vs 0.039

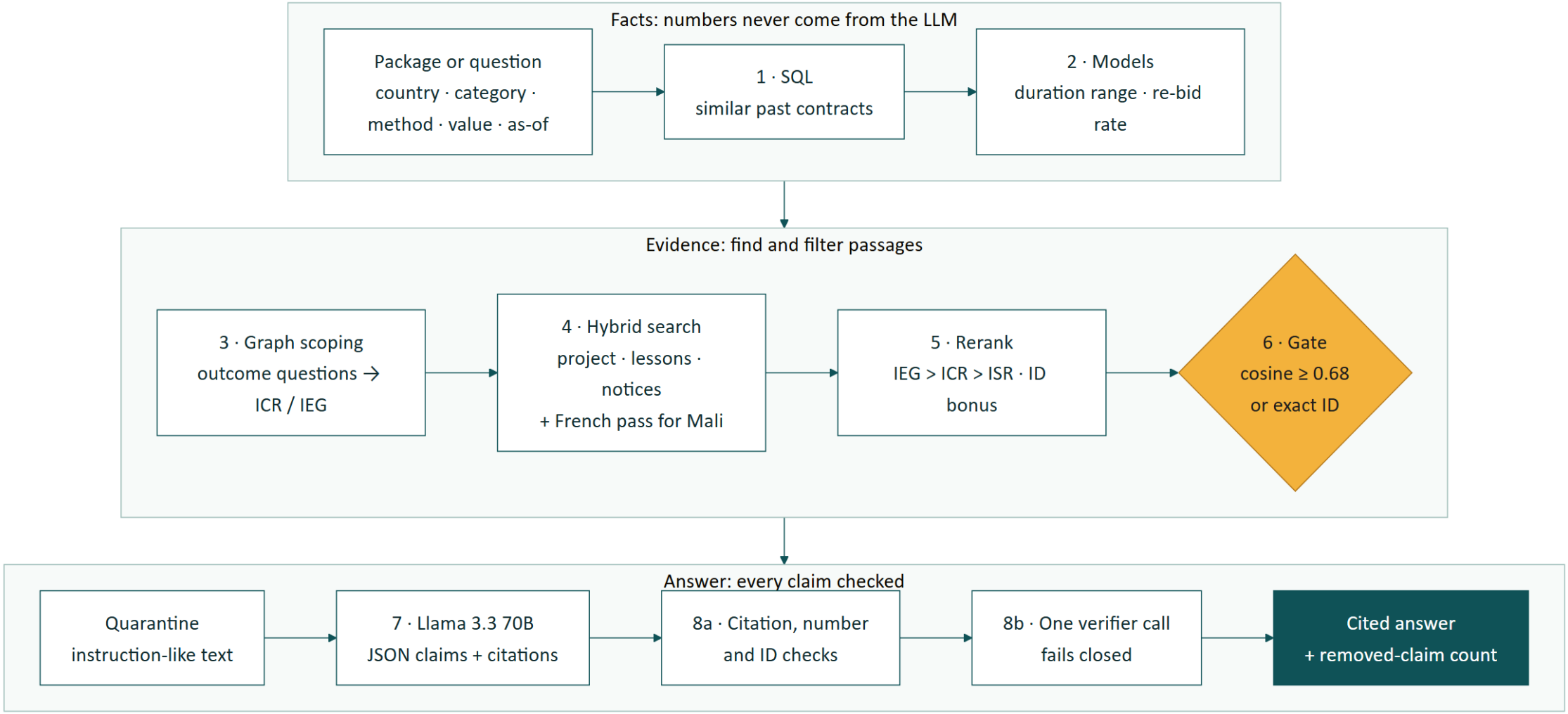
Brier score: grouped rates beat one global rate on the latest period

AUC 0.46

classifier not kept: rates with 90% intervals instead

Stated limitation one project holds 79% of positives; recent rates are a lower bound (re-advertisement takes a median 134 days).

Plan: a cited brief, built step by step

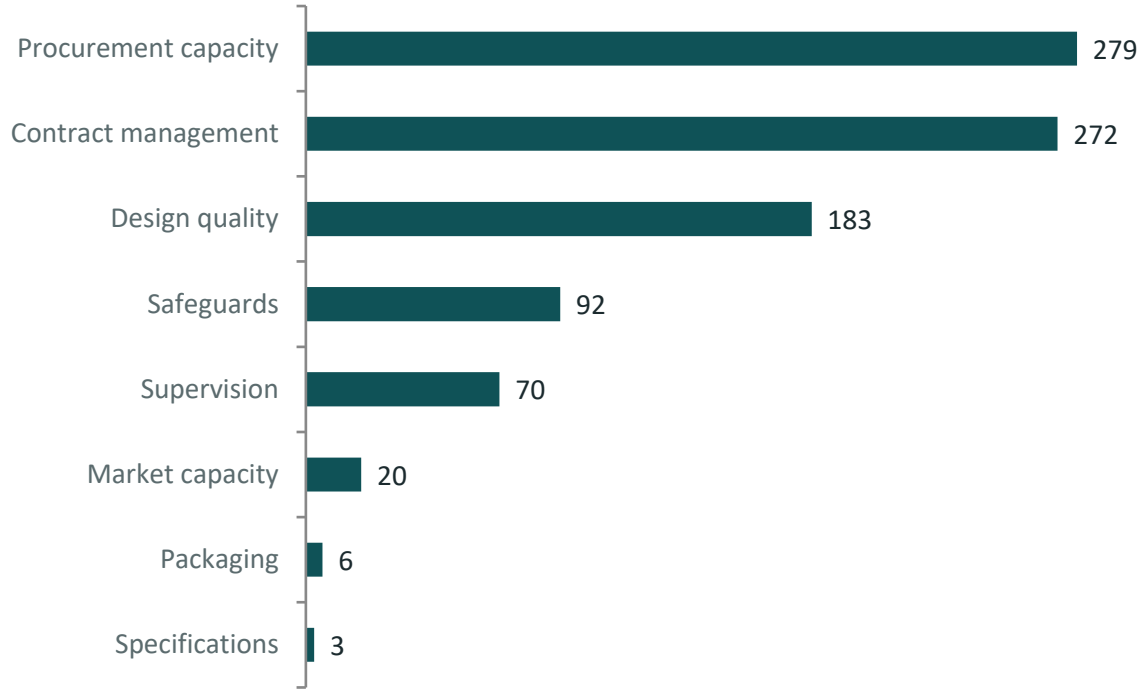


Tender design: learn from past notices

1 Plan · 2 **Tender design** · 3 Monitor · 4 Learn

- Past REOI / IFB text for similar packages: scope, lots, qualification criteria (4,338 notice passages)
- PAD procurement sections: planned arrangements and risks
- ICR and IEG lessons with verbatim quotes, by theme
- Ask about a reference or project ID and it is matched exactly

Lessons extracted from reports, by theme (first extraction run)



Stated limitation standard bidding documents, ToRs and bid evaluation reports are not public data. Tender-design recall@5 is 0.33 strict / 0.67 lenient on only 3 test questions.

Every claim is checked before it is shown



1.00
citation validity

0.86–0.905
claim support after guardrails (range over evaluation runs)

97.5–100%
of 40 kept claims supported, independent judge

3 / 3
off-topic questions refused

Mechanical

Each claim cites a supplied source; every number ≥ 10 and every project ID appears in the cited text

Verifier

One batched LLM call reads each claim beside its sources; unsupported or no verdict → removed

Quarantine

Passages with instruction-like text never reach the model (planted injection: blocked)

Conflicts

Final outcome: IEG > ICR > latest ISR; every rating shown with its date; flags raised

Monitor: early warnings that say “check”



91

tenders past their predicted P80 with no contract

16

disbursement-lag flags

38

active projects flagged

1 · 2

IEG downgrade · ISR volatility conflicts

Each flag shows its evidence

- Tender: notice date, predicted P50 / P80, days overdue
- Project: disbursement vs similar projects, latest ISR ratings
- Reported problems: amendments, delays, extensions with quotes

Stated limitation an overdue flag means “check”, not “late”: only about a third of contracts join to a notice, so a missing contract may be a join failure. The backtest has only 3 projects; the amendment indicator does not track extracted events ($r \approx -0.08$).

Evaluation, measured honestly

Test half, 26 questions	Before	After translation	Final
Recall@5 strict (gold passage)	0.52	0.52	**0.57**
Recall@5 lenient (any passage with the answer)	0.78	0.78	**0.83**
French-source questions (3)	0	0	0.33
Claim support after guardrails	–	0.905	0.86–0.893
Refusal accuracy	3 / 3	3 / 3	3 / 3

52 frozen questions, halves split by project; the gate is calibrated on one half, metrics reported on the other. Claim support varies a little between runs (LLM run-to-run variance).

Gate bug found and fixed

Vector Search scores $1/(1+L2^2)$, not cosine: the gate blocked 20 of 23 answerable questions. Now converted to cosine.

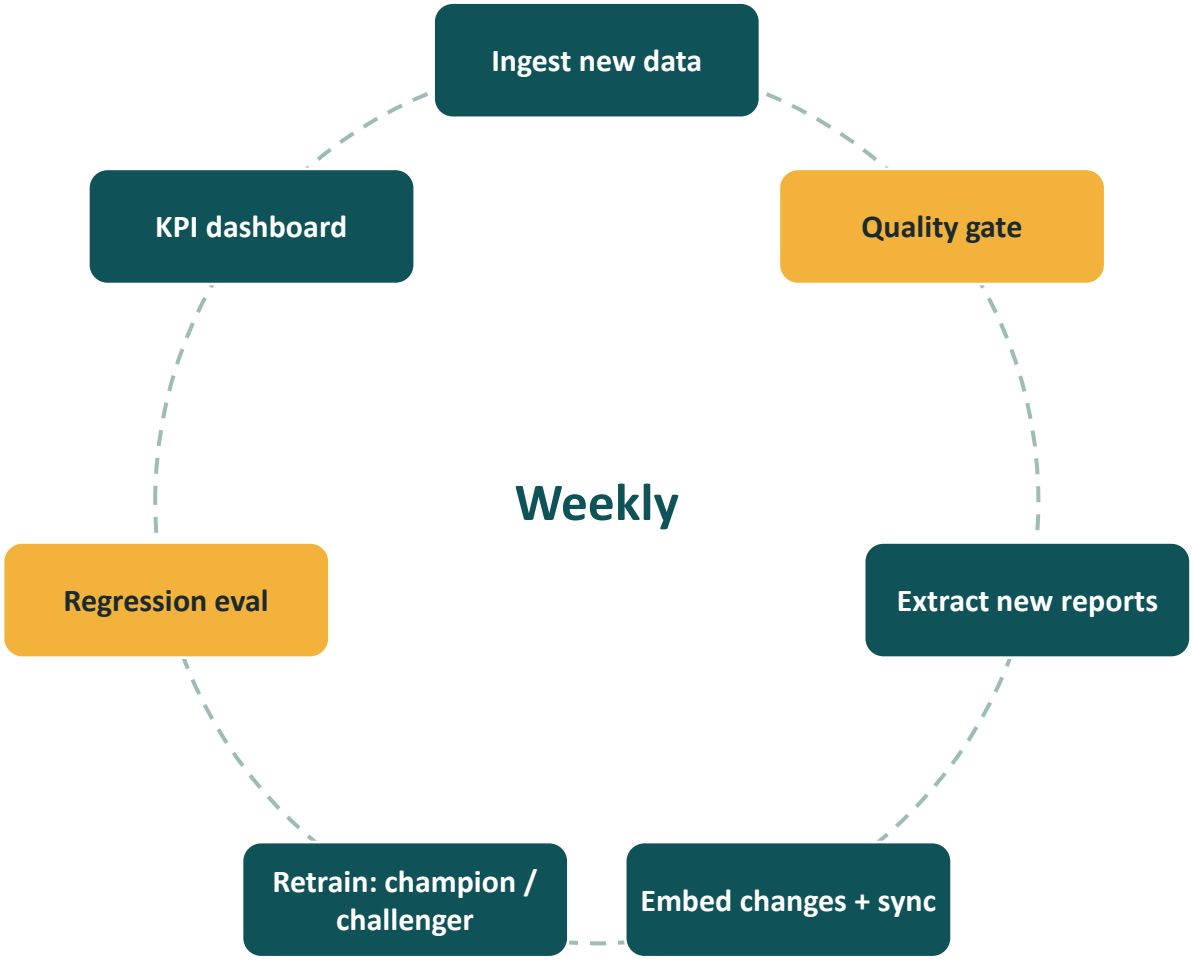
The gate is a weak filter

On- and off-topic scores overlap (0.66–0.86 vs 0.66–0.70). Refusals rest on the answer step and the verifier.

French is only partly solved

1 of 3 French questions found after English keywords + a French query pass. Fix: multilingual embeddings.

Learn: the loop runs every week



- Incremental at every step: only new documents are parsed, only changed passages re-embedded (e.g. 196 on 23 Oct)
- A model is promoted only if no worse on the latest time slice; the decision is logged in MLflow
- The run fails if recall@5 drops by more than 5 points
- 👍 / 👎 and “wrong source” feedback → reviewed → added to the evaluation set
- AI/BI dashboard: data health, model quality, assistant quality, usage

Rollout with go / no-go criteria



0	**Offline (this work)** Roads, 5 countries, frozen evaluation set	recall@5 ≥ 0.7 · claim support ≥ 0.9 · P50 beats baseline · P80 0.7–0.9
1	**Shadow pilot, 3 months** ~10 procurement specialists, one region, advisory only	≥ 60% weekly active · ≤ 5% wrong-source · 0 unsupported claims in a 100-answer audit
2	**Expansion** All transport, then energy and water; all countries	Metrics hold per sector · weekly refresh ≤ 4 h
3	**Integration** Links into STEP planning; internal data under access control	Security and data-governance sign-off

Phase 0 status (measured): P80 coverage 0.81 ✓ · claim support 0.86–0.905 (around the 0.9 target) · P50 ≈ baseline · recall@5 0.57 strict ✗ / 0.83 lenient ✓

What is not solved yet, and what comes next

Limitations (stated, not hidden)

- Small evaluation set (26 test questions); verifier and judge are the same model family
- Duration P50 only level with the baseline; governance features did not help
- No public amendment labels: the indicator is a heuristic
- French retrieval partly solved (1 of 3); English-only embeddings
- Contract ↔ passage links are sparse (1.3%)
- Free Edition limits: 3 apps, 24 h app runtime, a daily compute quota

Next steps

- Multilingual embedding model on Databricks
- STEP internal data: estimates, amendments, bid evaluations
- Larger, reviewed evaluation set; a different-family judge
- Shadow pilot with 10 specialists, then measure cycle time

Every number in this deck is measured; the tests, the evaluation and the app run the same pipeline.

Backup: supplier entity resolution

322,655 → 272,271

supplier records → suppliers

119 → 40

largest cluster, after the six guards

43–47 / 50

fuzzy-rule precision on a random sample (assistant-judged; spot-check pending)

R1	Same supplier ID
R2	Same cleaned name + country
R3	Near-identical name + country, firms only

Guards: shared-ID, common-name, numbers, joint venture, distinctive word, chain length. Masked names (“INDIVIDUAL CONSULTANT” with no ID) are never merged and get NULL history, never 0. Precision is preferred over recall: a wrong merge shows a user someone else's track record.

Backup: extraction from reports

1,548

LLM calls, approved in advance (dry run first), 0 errors

1,198

lessons kept, each with a verbatim quote

1,086

events kept: delays 344, amendments 223, cancellations 194, extensions 170

- Only events that HAPPENED: conditions, plans and project names are rejected (89 rejected as not happened)
- Quotes must appear word for word in the passage (83 event quotes rejected)
- A filter bug (“résili” matched “resilience”) cut the calls from 1,767 to 1,548 before any spending
- Ratings by regular expression, not LLM: 230 ratings for 16 projects
- Event → contract links: 3 of 1,086 (reports describe packages, not contracts)

Backup: the fake 0.93 AUC

0.93

classifier AUC on the first re-bid labels

120 of 171

positives were false

0.46

AUC after fixing the labels: classifier not kept

- “Projet de Relance et de Développement…”: relance means recovery, a project name, in 86 of 88 tenders
- Lebanon: “Not awarded lots… will be subject to re-bidding” printed in every invitation (34 of 43)
- Fix: wording inside a project name or after a future/conditional phrase never counts; recall re-checked on 125 notices
- Lesson: a high AUC on rare labels is a reason to read the positives, not to celebrate